SRAF-Net: Shape Robust Anchor-Free Network for Garbage Dumps in Remote Sensing Imagery

Xian Sun, *Senior Member, IEEE*, Yingfei Liu, *Student Member, IEEE*, Zhiyuan Yan, *Member, IEEE*, Peijin Wang, *Student Member, IEEE*, Wenhui Diao, *Member, IEEE*, and Kun Fu, *Member, IEEE*

***Abstract*— The detection of garbage dumps is of great significance for environmental protection. Recently, deep learning algorithms have brought impressive improvements for regular object detection. Different from conventional objects, garbage dumps are more inconspicuous and irregular and have the problem of blurred boundaries. To solve these problems, we propose a shape robust anchor-free network (SRAF-Net) that consists of feature extraction, multitask detection, and postprocessing. First, our network leverages the context-based deformable (CBD) module to combine context attention and deformable convolution. The contextual information obtained by context attention enables the network to focus on objects with inconspicuous appearance, while the deformable convolution enhances the feature representation. Then, we propose a multitask detection head to regress irregular garbage dumps in a more accurate and efficient way. The anchor-based methods need to define some anchors with a fixed shape. However, our detection method is anchor-free that learns the shapes of objects from training data. The detection head adaptively generates various shapes of bounding boxes with their classification confidences and localization confidences. Weighted by the localization confidences, we merge bounding boxes during postprocessing, which alleviates the blurred boundaries. In addition, we build a new public data set named garbage dumps data set (GDD) to verify the effectiveness of our method. Extensive experiments on GDD indicate that our method surpasses the existing detection methods in terms of speed and accuracy for the garbage dumps detection task.**

***Index Terms*— Anchor free, contextual information, deformable convolution, irregular object detection, remote sensing imagery.**

Manuscript received April 15, 2020; revised July 10, 2020 and August 29, 2020; accepted September 4, 2020. Date of publication October 5, 2020; date of current version June 24, 2021. This work was supported by the National Science Fund for Distinguished Young Scholars under Grant 61725105. *(Corresponding author: Xian Sun.)*

Xian Sun, Yingfei Liu, Peijin Wang, and Kun Fu are with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the Key Laboratory of Network Information System Technology (NIST), Institute of Electronics, Chinese Academy of Sciences, Beijing 100190, China, also with the University of Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China (e-mail: sunxian@mail.ie.ac.cn; yingfei@tju.edu.cn; wangpeijin17@mails.ucas.ac.cn; fukun@mail.ie.ac.cn).

Zhiyuan Yan and Wenhui Diao are with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Institute of Electronics, Chinese Academy of Sciences, Beijing 100190, China (e-mail: yanzy@aircas.ac.cn; dwh1031@gmail.com).

Digital Object Identifier 10.1109/TGRS.2020.3023928

I. Introduction

THE disposal of domestic garbage in developing countries has always been an important subject and deserves more attention with the increasing environmental pressures. Mixed garbage is often piled on the ground, causing serious pollution to the surrounding water, soil, and air. Therefore, it is important to monitor the location and expansion of garbage dumps.

Site-investigation is the main way to monitor garbage dumps. Since January 2017, the Ministry of Housing and Urban-Rural Development of China (MHURC) has been surveying garbage dumps reported by residents. However, the investigation of garbage dumps is time-consuming and laborious. Recently, with the development of optical remote sensing imaging technology, a growing repository of high-resolution images is available that shows excellent potential for large-scale object monitoring. Meanwhile, image processing using deep learning algorithms has gained momentum. Many works [1]–[9] proposed a series of approaches for object detection in remote sensing imagery and achieved excellent results on some rigid targets (e.g. airplanes, ships, bridges, and vehicles). Chen *et al.* [1] detected vehicle based on sliding windows and deep neural network. Long *et al.* [2] proposed a bounding-boxes regression algorithm for aircraft detection in remote sensing images. Wang *et al.* [6] introduced contextual information to detect small objects and achieved a tradeoff between the speed and accuracy on the DOTA [10] data set. These efforts greatly improved the efficiency of detecting rigid objects. However, different from these objects, garbage dumps have more irregular shapes, inconspicuous appearance, and blurred boundaries. As illustrated in Fig. 1, the shapes of garbage dumps are different from each other. Besides, garbage dumps in some scenes, such as woodland and farmland, have similar textures to their surroundings, resulting in the poor feature representation. Affected by the resolution, some blurred pixels at the edges in Fig. 1(c) and (d) will influence the detection performance, which is the problem of blurred boundaries.

There are a few algorithms [11]–[16] proposed for irregular objects in natural scenes. Deformable convolutional network (DCN) [15] is a typical method. As shown in Fig. 2(a), DCN [15] introduces deformable convolution in the backbone, which significantly improves the capacity of modeling

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

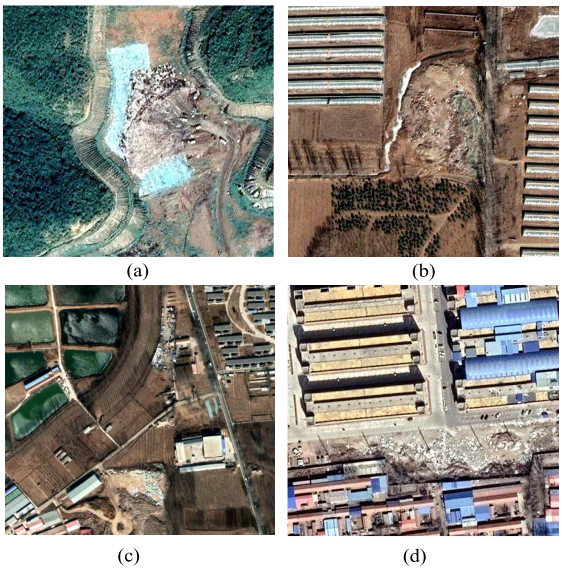


Fig. 1. Visualization of some garbage dumps in different scenarios. (a) Woodland. (b) Farmland. (c) Suburbs. (d) Town. From these images, we can find that garbage dumps are inconspicuous and irregular and have the problem of blurred boundaries.

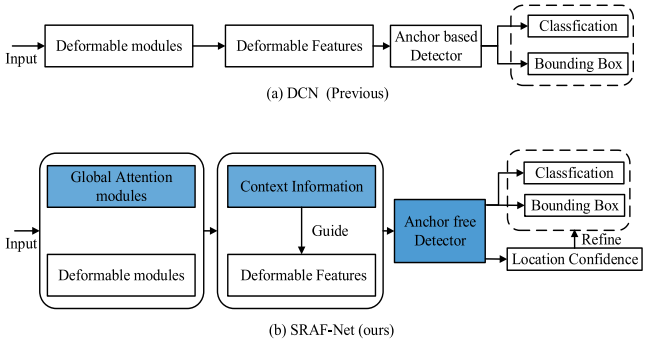


Fig. 2. Illustration of the detection pipeline. (a) DCN is an anchor-based detector. It introduces deformable convolution into the backbone to improve the capacity of modeling irregular objects in natural scenes. (b) SRAF-Net is an anchor-free detector. It uses the context information obtained by attention modules to enable the network to focus on the objects with poor feature representation. Localization confidences are the additional output for the correction of bounding boxes.

nonrigid objects. Then, it uses an anchor-based detector to regress the objects. DCN [15] achieves a better result on MS COCO [17] data set, especially some images contain leftovers (e.g. sandwiches, apples, and bananas) and distorted objects. However, it is a more challenging task to detect garbage dump in remote sensing images. First, DCN [15] predicts the bounding boxes by extra offset regression to the predefined anchor boxes. Compare with conventional objects in remote sensing images, it is difficult to describe garbage dumps with a specific shape and scale. The predefined shapes of anchor boxes are difficult to cover diverse garbage dumps, which affects the detection performance. Second, for nonrigid objects, although geometric constraint information can be obtained by deformable convolution, it is quite difficult to capture the appearance features that play a more important role in garbage dumps detection. Third, compared with objects with clear boundaries in natural scenes, the garbage dumps in remote sensing images have blurred boundaries. Thus, it is not an easy task to distinguish garbage dumps from surroundings.

To tackle these problems, we introduce shape robust anchor-free network (SRAF-Net), a novel object detection framework, which regresses garbage dumps in a robust and efficient way. As illustrated in Fig. 2(b), SRAF-Net combines the context attention and deformable convolution to extract features of garbage dumps. The context attention enables the network to focus on garbage dumps that have an inconspicuous appearance. With the guidance of context information, deformable convolution performs further feature extraction on irregular garbage dumps. The traditional regression has been replaced by a multitask anchor-free detector, which generates various bounding boxes adaptively based on training data. The additional branch in the detection head is designed for the prediction of localization confidences. Then, the localization confidences are used as the weight to merge multiple bounding boxes during postprocessing, which greatly alleviates the blurred boundaries. Our contributions can be summarized as follows.

1) We present a unified framework called SRAF-Net, which is implemented by a multitasking anchor-free detector. The model adaptively generates bounding boxes with various shapes, which is more accurate and effective for garbage dumps detection in high-resolution remote sensing imagery.
2) A context-based deformable (CBD) module is adopted to enhance the performance of feature extraction, through the combination of context attention and deformable convolution. CBD is more powerful than other structures to extract the features of objects with inconspicuous appearance and irregular shapes.
3) Merge Nonmaximum suppression (NMS) (Merge NMS) is proposed in postprocessing, which merges the bounding boxes and performs NMS. The fused bounding boxes alleviate the blurred boundaries and more accurate.

We build garbage dumps data set (GDD) referring to MHURC and conduct experiments on it. The data set will be published later. We hope our data sets and methods can reduce the workload of the garbage dumping investigation. The remainder of this article is organized as follows. In Section II, we briefly introduce some related methods. Then, we explain the details of our method in Section III and show the experiments in Section IV. Finally, Section V concludes this article with a discussion of the results.

II. Related Works

In this section, we briefly review some related works. The framework of detection, feature extraction method, and postprocessing algorithm are closely related to the performance of detection. Thus, we will introduce the related works from three corresponding aspects: framework of detection, feature extraction network, and NMS.

A. Framework of Detection

In recent years, object detection based on deep learning has developed rapidly. Most detectors can be divided into

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

anchor-based [18]–[25] methods and anchor-free [27]–[32] methods. The anchor-based detectors inherit the ideas from traditional sliding-window. Ren *et al.* [19] proposed the region proposal network (RPN) to make object detection end-to-end. The anchor boxes in RPN can be viewed as predefined sliding windows or proposals. The bounding boxes are predicted by an extra offsets regression to anchor boxes. The anchor-free methods do not require manually designed anchor boxes. The networks directly predict the bounding boxes and have shown great vitality. Redmon *et al.* [26] predicted bounding boxes by the points near the center of objects. It provided a way for real-time object detection. Tian *et al.* [29] used more points in objects to predict the bounding boxes and achieve a better performance. Another family of anchor-free detectors, such as CornerNet [31] and CenterNet [32], detected a series of keypoints and grouped them to form the final bounding boxes.

With the rapid development of these mechanisms, many works [1]–[5] proposed a series of deep learning approaches for object detection in remote sensing images. Chen *et al.* [1] proposed a vehicle detection model based on sliding windows and deep neural network. Long *et al.* [2] designed an unsupervised score-based bounding boxes regression algorithm for aircraft detection in remote sensing images. These methods improve the accuracy of existing detection algorithms, but the scale and rotation problems have not been solved. In recent years, a great deal of works [6]–[9] have been focused on solving the abovementioned two problems. Wang *et al.* [6] used the single-stage detector and designed an atrous spatial feature pyramid (ASFP) module to get contextual information. The ASFP module has greatly improved the performance of small objects. Pang *et al.* [7] proposed the remote sensing region-based convolutional neural network (R^2CNN) to speed up the process in large-scale images. R^2CNN is a lightweight detection architecture that is efficient in computation and memory consumption. The objects in the remote sensing image have directions. SCRDet [9] added a rotating branch to the existing two-stage network. In addition to the coordinates of the bounding boxes, the RPN network additionally regresses the angles of the bounding boxes. It has achieved excellent results in the field of oriented detection.

However, the irregular shape is the main difficulty of garbage dumps detection rather than multiscale and multidirection. There are some anchor-based methods that detect irregular objects in natural scenes, including PSENet [13], PAN [14], and DCN [15]. Dai *et al.* [15] introduced the deformable convolution and achieve a higher performance. PSENet [13] and PAN [14] modeled the irregular text regions in natural scenes by instance segmentation. These anchor-based methods are inefficient for the regression of irregular objects due to the complicated networks.

B. Feature Extraction Network

Currently, most detection frameworks use the backbone, such as VGG16 [33] and ResNet [34], which are based on regular convolution. These backbones are insufficient in solving some specific problems. Many methods have been proposed to enhance the feature representation of the backbone.

The regular convolution is inherently limited to model geometric transformations because of the fixed geometric structures of CNN modules. Dai *et al.* [15] proposed the deformable convolution, which allows the network to change autonomously during the training process. The deformable convolution greatly exceeds regular convolution in modeling irregular objects due to the flexibility of the sampling position.

The contextual information describes the symbiotic relationships between an object of interest and its surrounding environment. It plays a significant role in detecting inconspicuous objects, but it is difficult to obtain as the stacked convolutions are hard to optimize. Some works of face detection, such as SSH [35] and Pyramid Box [36], added a feature enhancement module (FEM) after the backbone to expand the receptive field. FEM often consists of multiple atrous convolutions and slightly improves the detection results. The advent of nonlocal [37] has made it possible to model long-range dependencies via the self-attention mechanism. It consists of a context module for global information modeling and a transform module for feature transformation. Some works, including CCNet [38] and GCNet [39], gradually simplified the context module. CCNet [38] designed the crisscross attention module to model spatial attention, which is resource-saving. GCNet [39] simplified the nonlocal block by using query-independent attention for all query positions. Compared with nonlocal network [37], GCNet [39] is more efficient in acquiring spatial contextual information.

C. Nonmaximum Suppression

Since NMS [40] was proposed in edge detection, it has been an integral part of computer vision. It is widely used in multiple tasks, such as feature point detection and object detection. The process is recursive that the box has a maximum classification score will be retained at each iteration. However, classification confidences are misaligned with localization confidences. Then, the bounding boxes with better localization may be incorrectly eliminated during the NMS [40] and lead to poor performance.

Recently, some works [41]–[44] are proposed to improve the results of NMS [40]. Soft-NMS [41] decayed the scores of bounding boxes with higher overlap by a continuous function instead of eliminating them directly. Thus, some valuable detection boxes with high scores will be retained after soft NMS [41], leading to a higher recall. IOU-guided NMS [42] replaced the classification confidences with the localization confidences by using a new prediction branch. Gaussian YOLO V3 [43] treated each generated bounding box as a Gaussian distribution to calculate the uncertainty of localization. Then, the classification confidences and uncertainty of localization are considered as the ranking keyword in NMS [40]. Furthermore, Softer NMS [44] used the standard deviation of the Gaussian distribution to merge bounding boxes, eliminating the blurred boundaries.

III. Methodology

A. Problem Setup

The task of garbage dumps detection can be viewed as a cascade of multilayer networks. For the layer i with input x_i,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

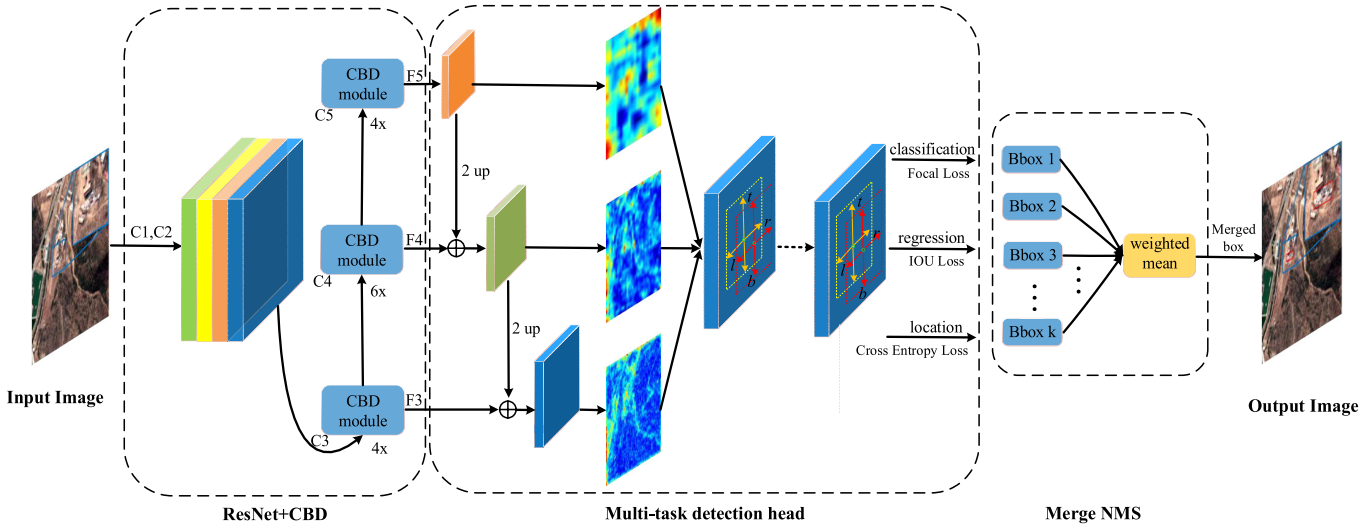


Fig. 3. Framework of SRAF-Net. It comprises three stages: ResNet with CBD module, multitask detection head, and merge-NMS. The CBD module is added to the ResNet to perform feature extraction on garbage dumps. The Multitask detection head is full convolution and anchor-free that learns the shapes of objects from training data gradually. The localization confidences are used as the weight to merge bounding boxes during postprocessing.

let q index a query element with content $x_{i,q}$ and k index a key element with content $x_{i,k}$. $W_{i,1}$ represents the weight of the layer. Then, the operation of convolution can be formulated as follows:

$$x_{i+1} = \sum_{k \in \Omega_q} \left[W_{i,1}^k D_i(q, k, x_{i,q}, x_{i,k}) \right]. \tag{1}$$

Here, the function D_i adaptively aggregates key contents in neighborhood Ω_q of position q. Both regular convolution and deformable convolution can be considered as special cases of (1).

Different from the calculation of convolution, the context attention often takes the entire feature map as input. The function A_i is used to calculate the attention map, and the input x_i will be multiplied by this map to obtain contextual information. Furthermore, an additional convolution operation is added to increase the nonlinearity of the network, and $W_{i,2}$ is the weight corresponding to it. The calculation of the attention layer is illuminated as follows:

$$x_{i+1} = \sum_{k \in \Omega_q} \left[W_{i,2}^k (A_i(x_i) \cdot x_i)_k \right]. \tag{2}$$

The context attention enables the network to focus on the region of interest, while the deformable convolution excels at modeling irregular objects; both are important for the process of detection. Thus, we use the regular convolution, deformable convolution, and context attention in our model. Besides, some postprocessing function ρ is necessary. The following equation can explain the entire process of detection framework with N-layer

$$\text{bboxes} = \rho \left(\prod_{i=1}^{N} \sum_{k \in \Omega_q} \left[W_{i,1}^k D_i(q, k, x_{i,q}, x_{i,k}) + W_{i,2}^k (A_i(x_i) \cdot x_i)_k \right] \right). \tag{3}$$

As shown in Fig. 3, our network is mainly composed of ResNet [34] with the CBD module, multitask detection head, and Merge-NMS. After using the CBD module, the network gradually improves the response of the object on the feature map. Multitask detection head is a full convolution network that detects objects with variant size by a three-layer feature pyramid (FPN [18]). Finally, the Merge-NMS is used as a postprocessing method to merge bounding boxes and perform NMS. Details will be described in the following section.

B. CBD Module

The conventional backbones, such as VGG [33] and ResNet [34], are slightly inadequate for the feature extraction as garbage dumps are irregular and inconspicuous objects. DCN [15] replaced 3×3 convolution with deformable convolution in the residual modulus. It improves the network's capacity to model irregular objects but still insufficient for inconspicuous objects.

The work of face detection proves that contextual information is a good enhancement for poor features. Inspired by it, we enhance $C_3 \sim C_5$ of ResNet50 [34] using CBD module that cascades the deformable convolution and context attention, as shown in Fig. 3. The context attention that we use is proposed in GCNet [39], which is lightweight and can effectively model contextual information. First, some regular convolutions in C_1 and C_2 are cascaded to extract shallow features. Then, the deformable convolution in CBD module gradually improves the response of the objects on the feature map, while the context attention enables the network pay more attention to regions of interest from the noisy environment. The entire pipeline of CBD module can be summarized as follows:

$$[F_3, F_4, F_5] = \prod_{i=1}^{N_1} \sum_{k \in \Omega_q} \left[W_{i,1}^k D_i(q, k, x_{i,q}, x_{i,k}) + W_{i,2}^k (A_i(x_i) \cdot x_i)_k \right]. \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

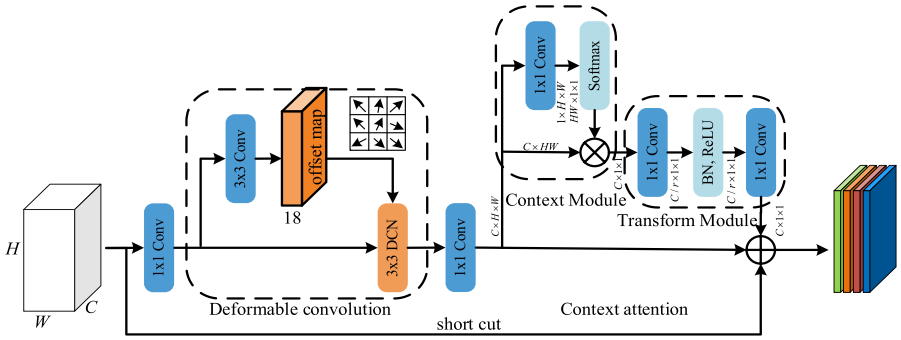


Fig. 4. Details of the CBD module. CBD module can be viewed as a combination of deformable convolution and context attention. Deformable convolution generates the offsets of 9 sampling points through a 3×3 branch. Context attention contains a context module and a transform module.

F_3, F_4, and F_5 are the feature maps generated by C_3, C_4, and C_5 of the backbone.

In 3×3 convolution, the number of spatial sampling point is $K = 9$, p_j is the offset of the corresponding point, and $p_j \in \{(-1,-1),(-1,0),\ldots,(1,1)\}$. For query element q and key element k on the feature map x_i, the function D_i can be calculated as follows:

$$D_i(q,k,x_{i,q},x_{i,k}) = \begin{cases} x_{i,k}, & \text{if } k = q + p_j \\ 0, & \text{else} \end{cases}. \tag{5}$$

As observed in Fig. 4, compared with regular convolution, the deformable convolution models the shape of objects by applying an additional offset Δp_j to the sample point p_j. The offset Δp_j can be obtained by an 3×3 convolution, which is lightweight and can be easily embedded into the original network. The weight of 3×3 convolution is $W_{i,3}$. Then, the deformable convolution can be expressed as

$$D_i(q,k,x_{i,q},x_{i,k}) = G_i\left(x_{i,k}, k, q + p_j + W_{i,3}^T x_{i,q}\right) \tag{6}$$

where G is a bilinear interpolation to calculate the value of the interpolation point.

The context attention in GCNet [39] simplified the nonlocal network [37], which models the spatial contextual information with less computation. It is lightweight that can be applied in multiple layers of the feature extraction network. Besides, it shares a similar structure with a squeeze-excitation network [45], which consists of a context module and a transform module, as shown in Fig. 4. In the context module, the features of $C \times H \times W$ are transformed through a 1×1 convolution and reshape its dimensions as $HW \times 1$. The attention map is obtained through the softmax layer, which reflects the importance of the query position in the feature maps. Then, the input features are reshaped as $C \times HW$, and multiply the attention map to produce contextual features. The calculation of the context module can be expressed as

$$A_i(x_i) = \sum_{j=1}^{HW} \frac{e^{W_{i,4} x_{i,j}}}{\sum_{m=1}^{HW} e^{W_{i,4} x_{i,m}}} \tag{7}$$

$W_{i,4}$ corresponds to the weight of 1×1 convolution in the context module. In the transform module, we deepen the network by two 1×1 convolutions that improve the expressiveness of contextual features. We set $r = 4$ to reduce the computational cost of the transform module. The contextual features are finally added to the features of each query position to form the output.

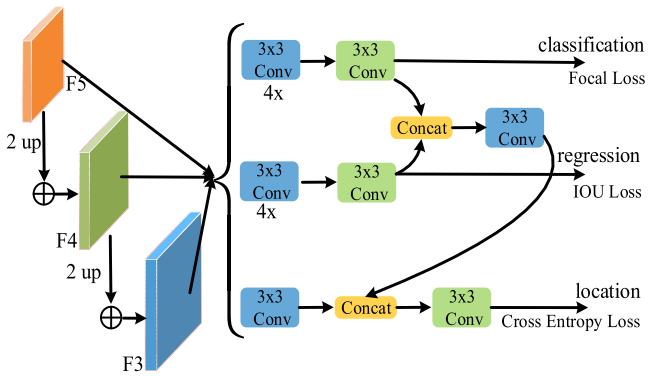


Fig. 5. Illustration of multitask detection head. The multitask detection head is a fully convolutional network and outputs additional localization confidences.

C. Bounding Boxes Regression With Multitask

The garbage dumps are nonrigid objects that have more complicated shapes than conventional objects. DCN [15] predicts the bounding boxes by applying two additional offsets to the predefined anchor boxes, which is inefficient compared with the single-stage algorithms [21]–[24]. At the same time, the performance of DCN [15] is sensitive to the sizes, aspect ratios, and the number of anchor boxes. A more powerful and efficient algorithm needs to be proposed to detect irregular objects.

Fig. 5 shows the multitask detection head that we proposed to detect garbage dumps. It is not only an anchor-free but also a single-stage method, which learns various shapes of objects adaptively during the training process. The detection head mainly comprises of regular convolution and can be expressed as follows:

$$[\text{Box}, \text{Cls}, \text{Loc}] = \prod_{i=1}^{N_2} \sum_{k=\in\Omega_q} \left[W_{i,1}^k D_i(q,k,x_{i,q},x_{i,k})\right]. \tag{8}$$

Box is bounding boxes we predicted, Cls is the classification confidences, and Loc is the localization confidences. As illuminated in Fig. 5, the model combines a three-layer feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

pyramid network to predict the bounding boxes. An additional localization branch is used to predict the intersection-over-union (IOU) between the detected boxes and ground-truth boxes. The predicted IOU values are considered as the localization confidences. Compared with the centerness branch used in FCOS [29], it is more reasonable to suppress the low-quality detected bounding boxes by localization confidences. In addition, the localization confidences can be used to solve the blurred boundaries, which we will introduce in the following section.

Especially, for an input image, the ground-truth bounding boxes can be defined as $\{B_i\}$,where $B_i = (x_0^i, y_0^i, x_1^i, y_1^i, c^i)$. Here, (x_0^i, y_0^i) and (x_1^i, y_1^i) denote the coordinates of the left-top and right-bottom corners of the bounding boxes, and c^i is the class it belongs to. For each location c^i on the feature map F, if the location (x, y) is in a ground-truth bounding box, it will be viewed as a positive sample and should regress this bounding box though (l^*, t^*, r^*, b^*). Here, (l^*, t^*, r^*, b^*) are the distances from the location to the four sides of the bounding boxes. They can be calculated as follows:

$$l^* = x - x_0^i, \quad t^* = y - y_0^i, \ r^* = x_1^i - x, \ b^* = y_1^i - y. \tag{9}$$

Combined with FPN [18], the model detects different sizes of objects on different levels of feature maps by limiting the range of bounding boxes regression for each level. The COCO [17] data set divides objects into three intervals according to $(0, 32, 99, \infty)$. Inspired by it, we used a three-level feature pyramid structure and set $(m_2, m_3, m_4, m_5) = (0, 32, 128, \infty)$ to detect small, middle, and large objects. For each location on all feature levels, we first compute the regression (l^*, t^*, r^*, b^*); if $m_{i-1} < \max(l^*, t^*, r^*, b^*) < m_i$, this sample will be assigned to regress the target in i_{th} level.

To solve the imbalance between positive and negative samples, we adopt the focal loss used in RetinaNet [24]. The classification loss is defined as

$$L_{\text{cls}}\left(p_{x,y}, p_{x,y}^*\right) = \begin{cases} -\alpha(1 - p_{x,y})^\gamma * \log(p_{x,y}), & p_{x,y}^* = 1 \\ -(1-\alpha) p_{x,y}^\gamma * \log(1 - p_{x,y}), & p_{x,y}^* = 0. \end{cases} \tag{10}$$

γ is usually greater than 1 to reduce the proportion of simple samples, while α is used to adjust the proportion of positive and negative samples. We set $\alpha = 0.25$ and $\gamma = 2$ as they are the default parameters in RetinaNet [24]. The localization confidences range from 0 to 1 that can be regarded as a probability, and we train it with a binary cross-entropy (BCE) loss

$$L_{\text{loc}}\left(l_{x,y}, l_{x,y}^*\right) = -l_{x,y}^* \log(l_{x,y}) - \left(1 - l_{x,y}^*\right) \log(1 - l_{x,y}). \tag{11}$$

For the regression of bounding boxes, we use the negative log IOU loss in UnitBox [25] as

$$L_{\text{reg}}\left(t_{x,y}, t_{x,y}^*\right) = -\ln\left(\frac{\text{Intersection}\left(\text{bbox}(t_{x,y}), \text{bbox}\left(t_{x,y}^*\right)\right)}{\text{Union}\left(\text{bbox}(t_{x,y}), \text{bbox}\left(t_{x,y}^*\right)\right)}\right). \tag{12}$$

Then, the total loss function that we used during the training process is that

$$\begin{aligned} L(p_{x,y}, t_{x,y}, l_{x,y}) = & \frac{\lambda_1}{N_{\text{all}}} \sum_{x,y} L_{\text{cls}}\left(p_{x,y}, p_{x,y}^*\right) \\ & + \frac{\lambda_2}{N_{\text{pos}}} \sum_{x,y \in GT} L_{\text{loc}}\left(l_{x,y}, l_{x,y}^*\right) \\ & + \frac{\lambda_3}{N_{\text{pos}}} \sum_{x,y \in GT} L_{\text{reg}}\left(t_{x,y}, t_{x,y}^*\right). \end{aligned} \tag{13}$$

L_{loc} and L_{reg} only calculate the loss for the positive samples in the ground truth. N_{all} and N_{pos} are constant used for normalization. λ_1, λ_2, and λ_3 are hyperparameters, and we set $\lambda_1 = \lambda_2 = \lambda_3 = 1$ to control the balance of different loss.

D. Merge NMS

Affected by the resolution, some blurred pixels in remote sensing imagery will influence the detection performance, which is the problem of blurred boundaries. Because of this factor, it is difficult to distinguish the garbage dumps from the surroundings accurately. Therefore, we design Merge NMS that mitigates the blurred boundaries during postprocessing. Let Box, Cls, and Loc are the bounding boxes, classification confidences, and localization confidences that output by SRAF-Net. The process of Merge NMS can be summarized as follows:

$$\text{bboxes} = \rho(\text{Box, Cls, Loc}|\theta). \tag{14}$$

The traditional NMS retains the bounding boxes with maximum scoring in every iteration. As a result, a significant number of bounding boxes are underutilized, which are also valuable. Merge NMS merges these boxes that should be suppressed. The detailed process is shown in Algorithm 1.

Algorithm 1 starts with a list of detection boxes (Box). The classification confidences (Cls) and localization confidences (Loc) are multiplied as the final scores (S). All bounding boxes will be sorted by scores S. In each loop, the box (b_m) with the highest score will be taken away from the set Box. Then, the boxes that have an overlap with b_m more than the threshold $\text{Merge}_{\text{thr}}$ are selected to merge. Formally, let x_m be a coordinate of b_m and x_k be the coordinate of the selected box at each loop. The boxes with high localization confidences l_k always have higher weights. The new coordinate of b_m is computed as follows, and it will be appended to the set of final detections D:

$$x_m = \frac{\sum_k l_k \times x_k}{\sum_k l_k}. \tag{15}$$

IV. Experimental Results

In this section, we present the experimental details of our SRAF-Net, including data sets, evaluation metrics, parameter settings, and experimental results.

A. Implementations

1) Data Set: Currently, there is a lack of data set related to garbage dumps or other irregular objects in remote sensing imageries. To apply object detection methods on irregular

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

TABLE I
STATISTICS FOR THE NUMBER OF GARBAGE DUMPS IN GDD

	train set					test set	
provinces	Anhui	Beijing	Fujian	Hainan	Shanxi	Guangdong	Tianjin
	155	245	61	50	479	135	128
images	990 (79.01%)					263 (20.99%)	
garbage dumps	1130 (79.08%)					299 (20.92%)	

Algorithm 1 Merge NMS Classification Confidences and Localization Confidences Are Multiplied as Final Confidences in the Algorithm. We Use These Confidences to Rank All Detected Bounding Boxes and Merge Them

Input: $Box = b_1, \ldots, b_n$, $Cls = c_1, \ldots, c_n$, $Loc = l_1, \ldots, l_n$, NMS_{thr}, $Merge_{thr}$. Box is $N \times 4$ matrix of detected bounding boxes. Cls and Loc map bounding boxes to their classification confidences and localization confidences respectively. NMS_{thr} is the NMS threshold. $Merge_{thr}$ is the threshold to merge.
Output: D, the set of detected bounding boxes with final confidences.

$D \leftarrow \emptyset$
$S = Cls \times Loc$
$T \leftarrow Box$
while $Box \neq \emptyset$
  $m = argmax(S)$
  $Box = Box - b_m$
  $idx \leftarrow IOU(b_m, Box) > NMS_{thr}$
  $Box = Box - Box[idx]$
  $idx \leftarrow IOU(b_m, T) > Merge_{thr}$
  $b_m \leftarrow \sum T[idx] \times Loc[idx] / \sum Loc[idx]$
  $D \bigcup \langle b_m, s_m \rangle$
end while

objects and monitor garbage dumps, we create the GDD that can be a benchmark for researchers to evaluate their algorithms.

To improve the quality of the data set, we use the results of site-investigation by MHURC since 2017, which includes seven provinces of China: Anhui, Beijing, Fujian, Guangdong, Hainan, Shanxi, and Tianjin. The data in this survey have precise latitude and longitude that dramatically reduces our workload. The large-scale aerial images that contain the garbage dumps were downloaded according to the latitude and longitude information. Each image is cropped to 1024×1024 pixels and has a resolution of 0.46 m. Then, we invite several experts to annotate the garbage dumps with (x, y, h, w). For each bounding boxes, (x, y) is the coordination of center point, and h and w represent the height and width, respectively.

Finally, we construct the data set GDD that includes 1253 images and 1429 garbage Dumps, covering various scenes, such as cities, villages, and mountains. As illustrated in Table I, we divide the data set into two parts (train set and test set) by province. Then, the train set is comprised of 990 images from Anhui, Beijing, Fujian, Hainan, and Shanxi, while the test set consists of 263 images from Guangdong and Tianjin.

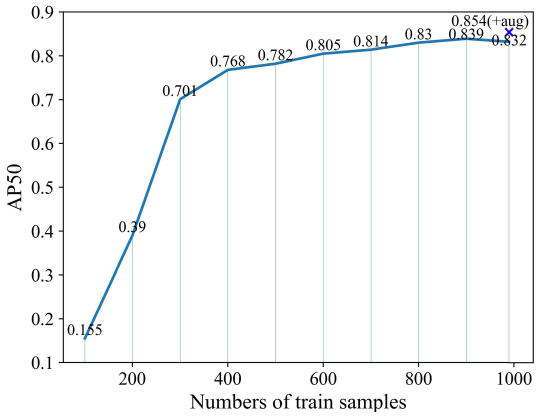


Fig. 6. Impact of the numbers of training samples. We use different numbers of training samples for training and record the results on the test set.

2) Evaluation Metrics: Considering the practical application of detecting garbage dumps, we adopt the standard average precision (AP) scores and frames per second (FPS) that are used in the MS COCO [17] data set. When evaluating the results of detection, if the predicted boundary box overlaps a ground-truth box more than a certain threshold, this box can be considered as a true positive (TP), otherwise, a false positive (FP). The ground-truth box has no matching predicted box that will be considered as a false negative (FN). Precision reflects the proportion of true positives in the results of detection, while recall demonstrates the proportion of the targets correctly recognized and detected. Precision and recall metrics are calculated as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{17}$$

In the postprocessing of detection, the precision and recall will change dynamically by setting different score thresholds, which is the precision–recall (PR) curve. The APs can be viewed as the areas under PR curves obtained by setting different overlapping thresholds. More specifically, AP50 is the area under the PR curve using the overlapping threshold of 0.50. In our experiments, we record AP50 and draw the PR curve corresponding to it. To assess the spatiotemporal complexity, we define the time for 100 km^2 (TFH) as a new metric. TFH represents the time required to process 100 km^2.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

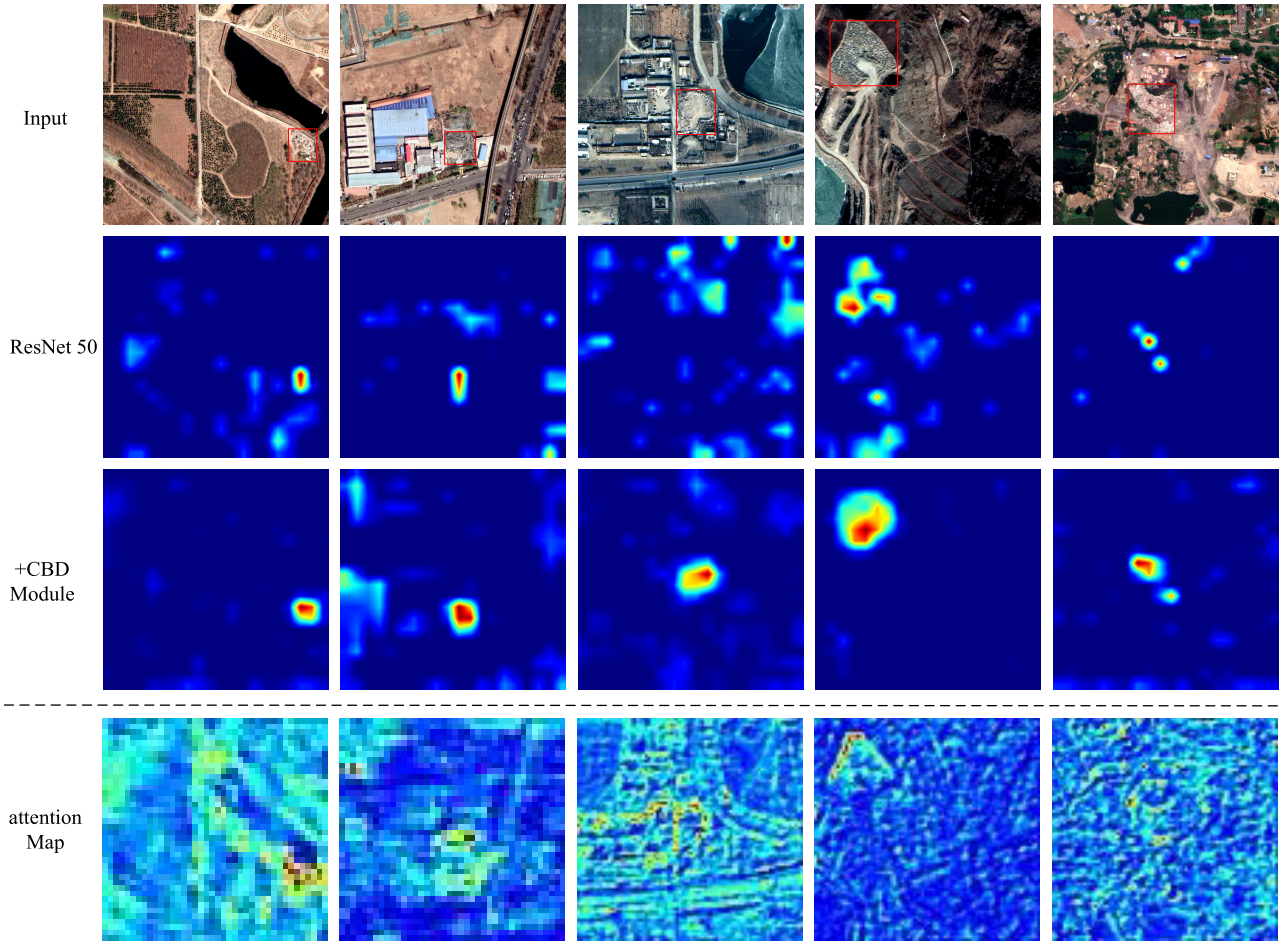


Fig. 7. First, we visual the feature map of ResNet and CBD modules through grad CAM. ResNet50 can only extract part of features, which is not enough for detection. Compared with ResNet50, the features are more comprehensive when the CBD module is added. Then, we visualized the heatmap of context attention. The attention map has a high response around the garbage dumps. When garbage dumps are hard to distinguish, some surrounding roads, farmlands, and residential areas also have higher response values. This environmental information is very helpful to the detection of garbage dumps.

It is obtained by dividing the running time (Times/s) and the space area (Areas/100 km^2). It is similar to FPS but more valuable in practical applications

$$\text{TFH} = \frac{\text{Times}}{\text{Areas}}. \tag{18}$$

3) Parameter Settings: The baseline of our method is a variant of FCOS [29] as the centerness branch is removed. ResNet50 [34] is the backbone of our model, and all layers, except for C_1 and C_2, are jointly finetuned with detection heads. More specifically, the model is trained with stochastic gradient descent (SGD) for 30 epochs, and every minibatch includes eight images. The initial learning rate is 0.001 and is reduced by a factor of 10 in the 20th epoch. Weight decay and momentum are set as 0.0001 and 0.9, respectively. We calculate the average of ten results to reduce fluctuations. Compared with other methods, the input images are resized to 1024 × 1024 for two-stage detectors and 640 × 640 for single-stage detectors. To be fair, the ResNet50 [34] is used as backbone for all models except for SSD [23] whose backbone is VGG16 [33]. The backbones of all models are pretrained on ImageNet. All experiments were carried out on a GTX1080 Ti GPU with CUDA10.0.

B. Evaluation of SRAF-Net

In this section, we comprehensively evaluate our method on the GDD data set. A series of ablation experiments show the effectiveness of our approach. Besides, our method tends to achieve better performance compared with methods for conventional objects and irregular objects.

1) Impact of the Data Volume: Some data sets for regular objects contain a large number of images and instances. DOTA [10] consists of 2806 images of 4096 × 4096 pixels, and NWPU VHR-10 [47] consists of 800 high-resolution remote sensing imageries. Compared with these data sets for regular objects, GDD has 1253 images of 1024 × 1024 pixels. To explore the impact of data volume, we gradually increase the number of training images, as shown in Fig. 6. The detection results only change little after the training samples reach 800. When using more data enhancement methods that used in SSD [23], the effect can continue to improve but is limited. These results indicate that GDD is big enough to train the one class detection model.

2) Ablation Experiments: We conduct a series of ablation experiments on the GDD data set.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

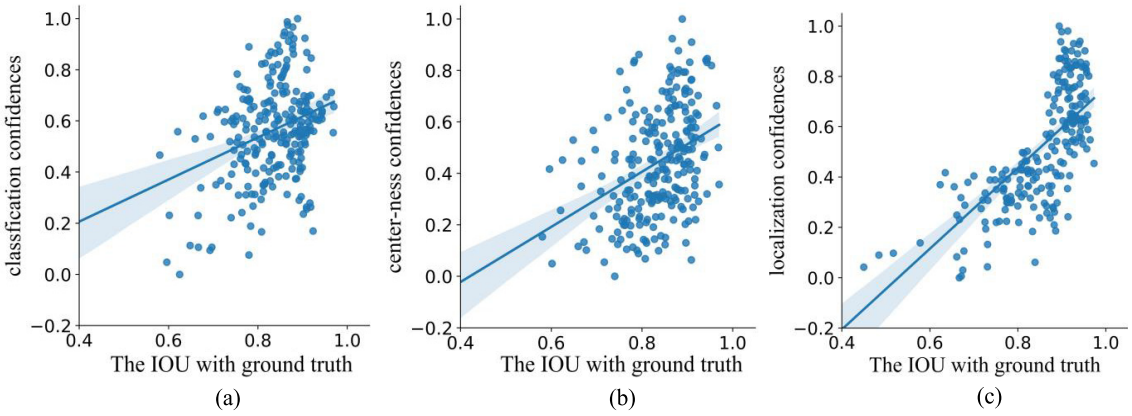


Fig. 8. Relevance to the quality of bounding boxes. (a) Correlation between classification confidences and the IOU truths. (b) Correlation between centerness and the IOU truths. (c) Correlation between localization confidences and the IOU truths.

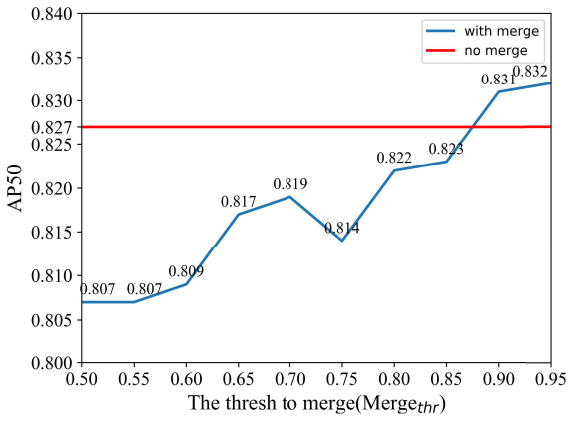


Fig. 9. Sensitivity analysis of hyperparameter Merge$_{\text{thr}}$. Merge$_{\text{thr}}$ ranges from 0.5 to 0.95.

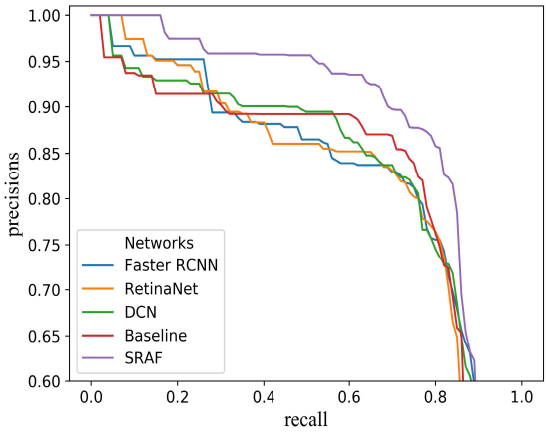


Fig. 10. PR curves of different methods. Faster RCNN and RetinaNet are conventional two- and single-stage algorithms, respectively. DCN is the algorithm for detecting irregular objects.

a) Effect of CBD module: Conventional convolutions are difficult to achieve good results in extracting features of inconspicuous objects. First, as shown in Table II, we add the context information and deformable convolution to $C_3 \sim C_5$ of ResNet50 [34], respectively, and both of them can achieve a better result. This shows that they are useful for feature enhancement of poor features. Then, we used four different addition methods to evaluate the CBD module that we designed, and all of them can achieve a certain improvement. When the CBD module is used in all $C_3 \sim C_5$ of ResNet50 [34], our methods exceed the baseline of 4.48%.

The grad CAM [46] is a useful tool for "visual explanations," and we use it for feature visualization. As shown in Fig. 7, ResNet50 [34] can only capture the local features of the object. Besides, due to environmental similarity, the extracted features contain more noise, which is harmful to the detection of garbage dumps. However, by using our CBD module, the network gets comprehensive information about garbage dumps. Then, we visualized the attention map of context attention to verify the motivation of context information. The attention map has a high response around the garbage dumps. When the garbage dumps are hard to distinguish, some context information plays an important role in detecting garbage dumps. Thus, the surrounding roads, farmland, and residential areas also have higher response values, as shown in Fig. 7.

TABLE II
ABLATION STUDY ON CBD MODULE

	C_3	C_4	C_5	AP50	ΔAP50
baseline	-	-	-	77.44%	-
Context	√	√	√	80.73%	+3.29%
Deform	√	√	√	79.26%	+1.82%
CBD	√	-	-	80.67%	+3.23%
CBD	-	√	-	79.57%	+2.13%
CBD	-	-	√	78.48%	+1.04%
CBD	√	√	√	**81.92%**	**+4.48%**

b) Effect of localization confidences: To prove the effectiveness of the localization confidences branch that we designed, we make a comparison with the centerness branch that is proposed in FCOS [29]. As shown in Fig. 8, the classification confidences and centerness confidences are not related to the quality of the bounding boxes. After using the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

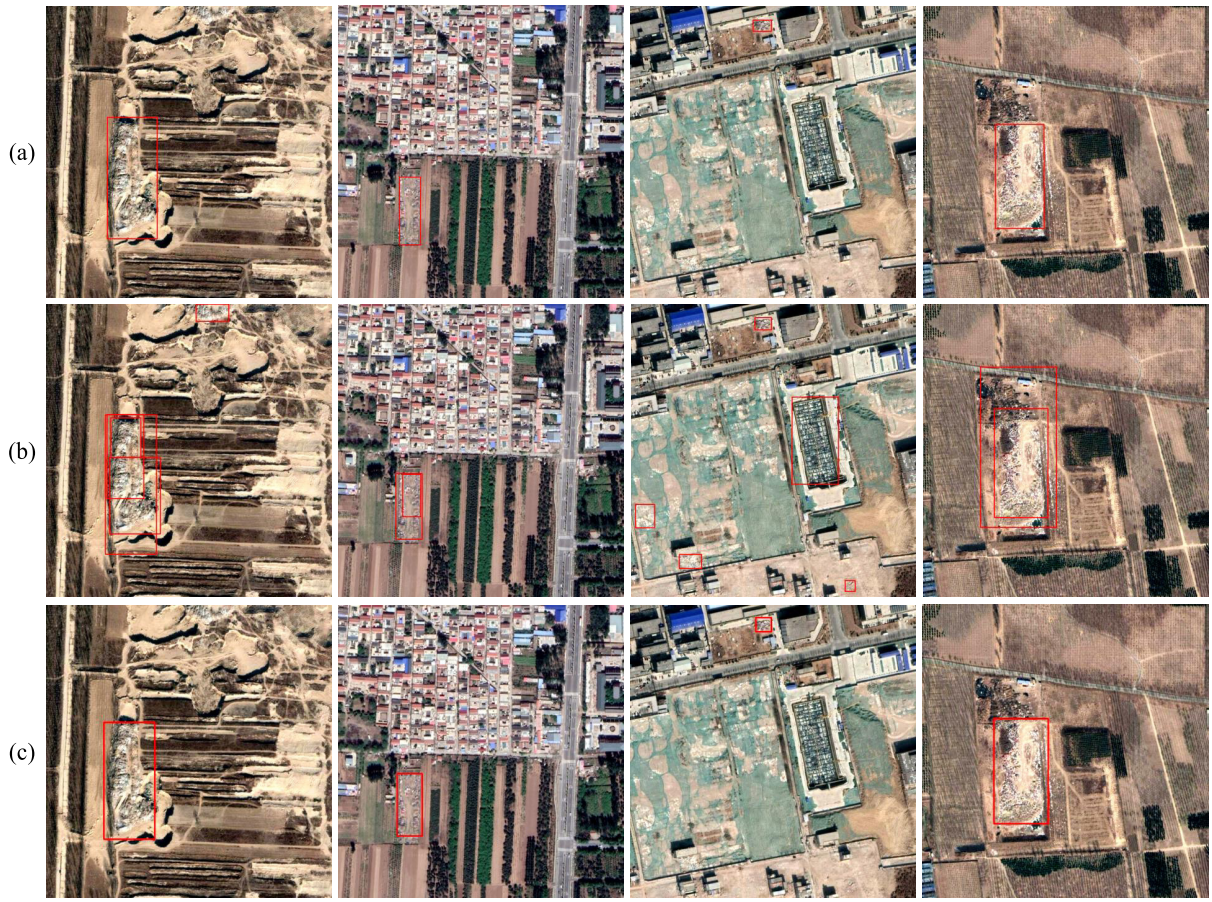


Fig. 11. Visualization of different methods. The bounding boxes in (a) are ground truths. (b) Detection result by DCN. (c) Detection result by SRAF-Net. SRAF-Net detects garbage dumps more accurate while maintaining a low false alarm rate.

localization confidences, the correlation is greatly improved. Similar to FCOS [29], we multiply the localization confidences and classification confidences as the final score. As shown in Table III, the AP50 is increased by 0.76% with the localization confidences branch. The localization confidences branch is very lightweight, with only a slight sacrifice in speed.

c) Effect of Merge NMS: As shown in the experiments in Table IV, we perform Merge NMS by classification confidences and localization confidences, respectively. All of them can improve the effectiveness of detection, which indicates that Merge NMS is beneficial for mitigating blurred boundaries. Compared with other fusion methods, the time overhead of Merge NMS is less than 10%.

$\text{Merge}_{\text{thr}}$ is a hyperparameter in Merge NMS, and the boxes will be fused when their IOU value greater than this threshold. As shown in Fig. 9, the setting of $\text{Merge}_{\text{thr}}$ is important to the Merge NMS. The detection result will be improved when $\text{Merge}_{\text{thr}}$ greater than 0.9. We think this is related to the problem of blurred boundaries. If we set a high threshold, the bounding boxes used for merging only have slight differences at the boundary. On the contrary, if we set a lower $\text{Merge}_{\text{thr}}$, the boxes used for fusion will have a large gap and influence the final results. Thus, $\text{Merge}_{\text{thr}}$ should be set to a high value.

TABLE III
ABLATION STUDY ON LOCALIZATION CONFIDENCES BRANCH

	centeness	localization	AP50	FPS(img/s)
baseline	-	-	77.44%	33.59
+CBD	-	-	81.92%	24.04
1	√	-	82.13%	23.05
2	-	√	**82.68%**	22.80

TABLE IV
ABLATION STUDY OF MERGE NMS

	classification	localization	AP50	FPS(img/s)
+CBD+Loc	-	-	82.68%	22.80
Merge	√	-	83.24%	21.37
Merge	-	√	**83.17%**	21.33

3) Comparisons With Other Methods: To validate the effectiveness of our method, we compare it with some classical algorithms for conventional objects and irregular objects in Table V. The conventional methods are slightly insufficient when detecting irregular objects, such as garbage dumps.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

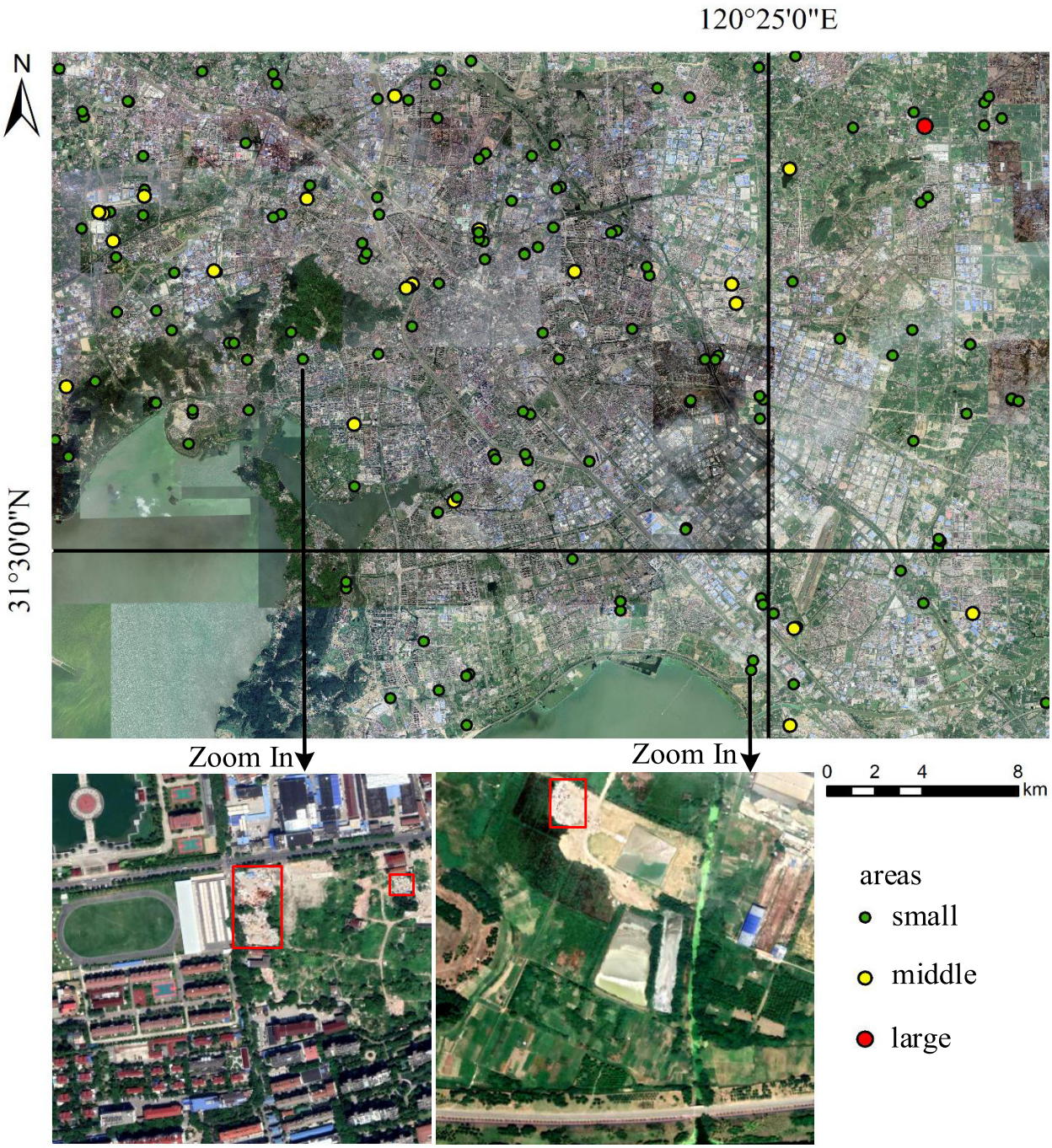


Fig. 12. Detection results of Suzhou, China. We visualize the results into three colors according to the area. The green points are garbage dumps small than 1000 m^2. The yellow points are garbage dumps range from 1000 to 10 000 m^2. The red points are garbage dumps big than 1000 m^2.

There are only minor differences in accuracy between the two- and single-stage algorithms. At the same time, the single-stage algorithms are more efficient than two-stage. When comparing RetinaNet [24] and FCOS [29], we find that the regression by anchor-free can bring some improvements in both accuracy and speed. Therefore, the single-stage algorithm based on anchor-free is more suitable for irregular objects. DCN [15] is one kind of algorithms for irregular objects that use the deformable convolution in Faster RCNN [19]. The performance of DCN [15] shows that the feature extraction of irregular objects plays a vital role in detection. Compared with these methods, our method achieves excellent performance in both speed and accuracy.

As shown in Fig. 10, we plot the PR curves of different algorithms, and our method has an improvement in both precision and recall. We visualized the detection results in Fig. 11, and our algorithm achieved better detection results on the garbage dumps that are inconspicuous and irregular. SRAF-Net can detect objects missed by other methods while maintaining a low false alarm rate. Therefore, our method solves the three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

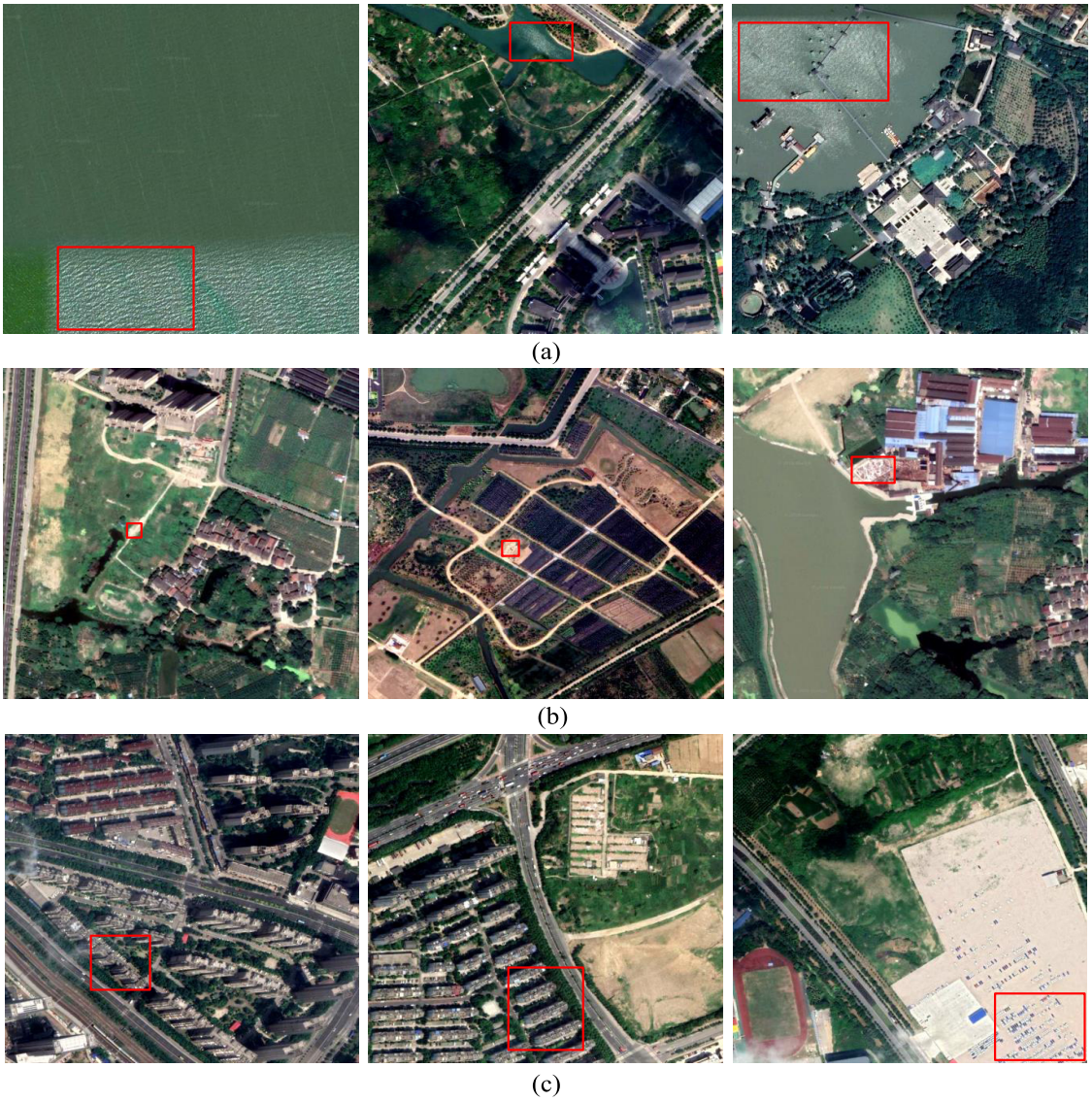


Fig. 13. Three types of errors in practical applications. (a) Some objects with noise are mistakenly detected as garbage dumps. (b) Some blurred objects are mistakenly detected as garbage dumps. (c) Some objects with complex textures are mistakenly detected as garbage dumps.

problems described earlier and is suitable for objects with the poor features.

C. Discussion

1) Generalization Performance: To verify the generalization performance of garbage dumps detection, we test our method on the remote sensing imagery of Suzhou, China. The image has $70\,400 \times 49\,408$ pixels covering an area of 736 km^2. An appropriate score threshold is important, and we set different score thresholds on the GDD data set and record multiple sets of precision and recall in Table VI. Since the cities of the train set and the test set are different, Table VI can provide a reference for the choice of score threshold. Higher thresholds reduce the number of misclassifications and achieve higher precision, while lower thresholds allow the acceptance of more boxes and achieve a higher recall. Finally, we choose 0.3, where both precision and recall are kept at a high level. If we consider applying the whole framework to other data sets, we think it is necessary to carry out some experiments to make the algorithm more accurate. To improve efficiency, we only need to extract a small amount of data for the experiment. Once the parameter is selected, it can be applied to the whole data set. Therefore, the experiment consumes very little resources.

Due to the efficiency of our model, the entire test can be completed in just 3 min. The massive and complex backgrounds cause serious false alarms. We filter the 1168 detection results by one staff and get 162 garbage dumps in Fig. 12. Coupled with this part of the time, we can complete the investigation of a city in only one hour, which significantly improves the efficiency compared with field investigations.

2) Error Analysis: To analyze the problems in practical applications, we visualized some detection results in Fig. 13. The wrong detections mainly caused by three factors: 1) the noise in some images; 2) blurred and small-sized objects; and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISONS WITH OTHER METHODS

Objects	Methods	AP50	FPS(/s)	TFH(s)
Conventional	Anchor-based two-stage			
	Faster RCNN+FPN [19]	79.47%	12.93	302
	Cascade RCNN [20]	78.27%	9.61	406
	Anchor-based single-stage			
	SSD512 [23]	75.33%	25.83	151
	RetinaNet [24]	79.46%	24.33	161
	FMSSD [6]	73.35%	16.17	242
	Anchor-free single-stage			
	FCOS [29]	80.15%	28.80	136
irregular	Anchor-based two-stage			
	DCN [15]	80.80%	11.31	345
	Anchor-free single-stage			
	baseline	77.44%	33.59	116
	SRAF-Net	**83.24%**	21.37	183

TABLE VI
PRECISION AND RECALL AT DIFFERENT THRESHOLDS

Score threshold	Ground truths	Detected boxes	Recall	Precision
0.05	299	1460	0.933	0.191
0.1	299	748	0.913	0.365
0.2	299	473	0.890	0.562
0.3	299	381	0.860	0.675
0.4	299	328	0.846	0.771
0.5	299	286	0.799	0.836

3) objects with complex textures such as buildings and parking lots.

The first factor mainly appears on the surface of the water, which is more susceptible to the noise. We can filter detection results on the surface of the water to eliminate such errors. The second factor can be limited by using images with appropriate spatial resolution. The third factor is the hardest to resolve. Large-scale remote sensing imagery always contains a variety of scenarios. Some textures of complex scenes are similar to the garbage dumps that result in misclassification. This error is difficult to avoid and need to be manually cleared.

V. CONCLUSION

In this article, we propose SRAF-Net, a shape robust model for garbage dumps detection in remote sensing imagery. The garbage dumps have inconspicuous appearances, more irregular shapes, and blurred boundaries than conventional objects. Thus, it is not an easy task to detect garbage dumps. Compared with conventional methods, SRAF-Net is an anchor-free algorithm where the regression of irregular objects is achieved in a more powerful and efficient way. The CBD module combines deformable convolution and context attention to extract features on inconspicuous and irregular garbage dumps. Merge-NMS uses the localization confidences obtained in the detection head to merge bounding boxes and reduces blurred boundaries. To verify the effectiveness of our method, we build the GDD data set for the detection of garbage dumps. The prediction results on the GDD data set demonstrate that our method has more advanced performance than other detection methods. In the future, we plan to further reduce the false alarm rate and improve the accuracy of regression. Some automated methods for parameter setting will be adopted to make the algorithm more flexible. Besides, we will use the multitemporal applications to capture the long-term trends of garbage dumps.

REFERENCES

[1] X. Chen, S. Xiang, C.-L. Liu, and C.-H. Pan, "Vehicle detection in satellite images by hybrid deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1797–1801, Oct. 2014.

[2] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.

[3] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[4] W. Diao, X. Sun, X. Zheng, F. Dou, H. Wang, and K. Fu, "Efficient saliency-based object detection in remote sensing images using deep belief networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 2, pp. 137–141, Feb. 2016.

[5] J. Yan, H. Wang, M. Yan, W. Diao, X. Sun, and H. Li, "IoU-adaptive deformable R-CNN: Make full use of IoU for multi-class object detection in remote sensing imagery," *Remote Sens.*, vol. 11, no. 3, p. 286, Feb. 2019.

[6] P. Wang, X. Sun, W. Diao, and K. Fu, "FMSSD: Feature-merged single-shot detection for multiscale objects in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3377–3390, May 2020.

[7] J. Pang, C. Li, J. Shi, Z. Xu, and H. Feng, "$\mathcal{R}^2$-CNN: Fast tiny object detection in large-scale remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5512–5524, Aug. 2019.

[8] R. Dong, D. Xu, J. Zhao, L. Jiao, and J. An, "Sig-NMS-based faster R-CNN combining transfer learning for small target detection in VHR optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8534–8545, Nov. 2019.

[9] X. Yang *et al.*, "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8231–8240.

[10] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[11] X. Zhou *et al.*, "EAST: An efficient and accurate scene text detector," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5551–5560.

[12] M. Jaderberg *et al.*, "Spatial transformer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 2017–2025.

[13] X. Li, W. Wang, W. Hou, R.-Z. Liu, T. Lu, and J. Yang, "Shape robust text detection with progressive scale expansion network," 2018, *arXiv:1806.02559*. [Online]. Available: http://arxiv.org/abs/1806.02559

[14] W. Wang *et al.*, "Efficient and accurate arbitrary-shaped text detection with pixel aggregation network," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8440–8449.

[15] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[16] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.

[17] T.-Y. Lin *et al.*, "Microsoft coco: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Zürich, Switzerland: Springer, 2014, pp. 740–755.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.

[18] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[19] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.

[20] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[21] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7263–7271.

[22] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*. [Online]. Available: http://arxiv.org/abs/1804.02767

[23] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Amsterdam, The Netherlands: Springer, 2016, pp. 21–37.

[24] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[25] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, "UnitBox: An advanced object detection network," in *Proc. ACM Multimedia Conf. (MM)*, 2016, pp. 516–520.

[26] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[27] L. Huang, Y. Yang, Y. Deng, and Y. Yu, "DenseBox: Unifying landmark localization with end to end object detection," 2015, *arXiv:1509.04874*. [Online]. Available: http://arxiv.org/abs/1509.04874

[28] C. Zhu, Y. He, and M. Savvides, "Feature selective anchor-free module for single-shot object detection," 2019, *arXiv:1903.00621*. [Online]. Available: http://arxiv.org/abs/1903.00621

[29] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," 2019, *arXiv:1904.01355*. [Online]. Available: http://arxiv.org/abs/1904.01355

[30] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi, "FoveaBox: Beyond anchor-based object detector," 2019, *arXiv:1904.03797*. [Online]. Available: http://arxiv.org/abs/1904.03797

[31] H. Law and J. Deng, "Cornernet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[32] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[33] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*. [Online]. Available: http://arxiv.org/abs/1409.1556

[34] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[35] M. Najibi, P. Samangouei, R. Chellappa, and L. S. Davis, "SSH: Single stage headless face detector," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4885–4894.

[36] X. Tang, D. K. Du, Z. He, and J. Liu, "Pyramidbox: A context-assisted single shot face detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 797–813.

[37] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[38] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[39] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," 2019, *arXiv:1904.11492*. [Online]. Available: http://arxiv.org/abs/1904.11492

[40] A. Rosenfeld and M. Thurston, "Edge and curve detection for visual scene analysis," *IEEE Trans. Comput.*, vol. C-20, no. 5, pp. 562–569, May 1971.

[41] N. Bodla, B. Singh, R. Chellappa, and L. S. Davis, "Soft-NMS—Improving object detection with one line of code," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5561–5569.

[42] B. Jiang, R. Luo, J. Mao, T. Xiao, and Y. Jiang, "Acquisition of localization confidence for accurate object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 784–799.

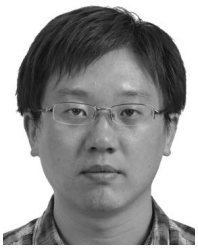
[43] J. Choi, D. Chun, H. Kim, and H. Lee, "Gaussian yolov3: An accurate and fast object detector using localization uncertainty for autonomous driving," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 502–511.

[44] Y. He, C. Zhu, J. Wang, M. Savvides, and X. Zhang, "Bounding box regression with uncertainty for accurate object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2888–2897.

[45] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[46] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[47] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.



Xian Sun (Senior Member, IEEE) received the B.Sc. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is a Professor with Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.



Yingfei Liu (Student Member, IEEE) received the B.Sc. degree from Tianjin University, Tianjin, China, in 2018. He is pursuing the master's degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include computer vision and deep learning, especially on object detection, semantic segmentation, generative adversarial networks, and remote sensing.



Zhiyuan Yan (Member, IEEE) received the B.Sc. degree from Xiamen University, Xiamen, China, in 2016, and the M.Sc. degree from Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2019.

She is an Assistant Engineer with Aerospace Information Research Institute, Chinese Academy of Sciences. Her research interests include computer vision and remote sensing image analysis.



Peijin Wang (Student Member, IEEE) received the B.S. degree from Tianjin University, Tianjin, China, in 2017. She is pursuing the master's degree with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

Her research interests include computer vision and deep learning, especially on object detection, semantic segmentation, and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.



Wenhui Diao (Member, IEEE) received the B.Sc. degree from Xidian University, Xi'an, China, in 2011, and the M.Sc. and Ph.D. degrees from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2016.

He is an Assistant Professor with Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image analysis.



Kun Fu (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 1995, 1999, and 2002, respectively.

He is a Professor with Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include computer vision, remote sensing image understanding, and geospatial data mining and visualization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:01:50 UTC from IEEE Xplore. Restrictions apply.